

Aman Samani

Full Stack Software Engineer

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Professional Summary

Full Stack Software Engineer (B.Tech CS, 2026) specializing in scalable web and mobile applications using the MERN stack. Skilled in designing production-grade REST APIs, adaptive algorithms, and secure authentication systems. Deployed Gallery Toons, a wallpaper platform used by 100+ students, and currently building Fitlip, an AI-powered health app with an adaptive recommendation engine. Strong focus on backend architecture and AI-enabled product features.

Technical Skills

Languages	JavaScript (ES6+), Python, Java
Core CS	Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks
Frontend	React.js, React Native (Expo), HTML5, CSS3
Backend	Node.js, Express.js, REST API Design, JWT Authentication, Middleware
Databases	MongoDB, PostgreSQL, SQL
Tools & DevOps	Git, GitHub, Postman, Vercel, Railway
Additional	Machine Learning (Scikit-learn, XGBoost, Pandas), REST/Third-Party API Integration

Projects

AI Personal Health Assistant — Fitlip

React Native · Node.js · MongoDB · Redis · Gemini API

- Building a cross-platform mobile app delivering AI-personalized nutrition and workout plans (in active development); backend previously ran live on Railway for 2 months during testing, now developed locally while adding features.
- Designed an adaptive calorie-adjustment engine that evaluates 7-day adherence and weight-change trends to auto-tune targets weekly, on top of a Mifflin-St Jeor BMR/TDEE and macro-split calculation core.
- Integrated Google's Gemini API for AI meal-plan generation with Zod schema validation and a post-generation correction algorithm that clamps AI output to calorie budgets, preventing overshoot from LLM inconsistency.
- Built a background job pipeline with BullMQ/Redis (retry + exponential backoff) to run scheduled weekly adjustments and trigger Expo push notifications.
- Implemented a full auth system with JWT, bcrypt password hashing, Google OAuth, and OTP-based password recovery via email.
- Designed **44+ REST API endpoints** across auth, nutrition, workout, and tracking modules, backed by a curated 300+ item Indian/international food database with normalized macros.

Fraud Detection — Transaction Classification Model

Python · XGBoost · scikit-learn

- Built an end-to-end ML pipeline to classify fraudulent transactions in a severely imbalanced dataset.
- Applied SMOTE oversampling to address class imbalance, then trained an XGBoost classifier (300 estimators) on the resampled data.
- Tuned the decision threshold to optimize F1-score, achieving **91.1% recall** and **80.5% F1-score** (72.1% precision) on a held-out test set of 465K+ transactions.
- Achieved a **0.995 ROC-AUC**, correctly identifying 479 of 526 fraudulent transactions in testing.

Gallery Toons — Wallpaper Web App

React · Vite · React Router · Live on Vercel

- Developed a responsive wallpaper platform with search, filtering, and one-click downloads.
- Grew to **100+ active users** through iterative UI/UX improvements and performance tuning.

- Optimized image delivery with WebP compression and native lazy loading, achieving a **90/100 Google Lighthouse performance score** (mobile, slow 4G) with zero layout shift.

Leadership & Achievements

- Led a 4-member cross-functional team (backend, frontend, UI/UX, research) as Project Lead for a final-year capstone, presented to faculty advisor Dr. Harsh Dev.
- Active problem solver on LeetCode in Java, with focus on arrays, strings, and dynamic programming.

Education

Babu Banarsi Das University, Lucknow

Sept 2022 – June 2026

B.Tech in Computer Science

GPA: 7.6 / 10.0